

		$\Delta x \leq (h)/(0.0100 \times 0.0004 \times 300)$ $= 5.53 \times 10^{-31} \text{ m}$
29	A	$eV = \frac{p^2}{2m}, \text{ giving } p = \sqrt{2meV}$ $\text{De Broglie wavelength } \lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$ $\text{Frequency} = c/\lambda = \frac{c\sqrt{2meV}}{h}$
30	C	Paper stops alpha particles, so $\alpha + \beta + background = 352 \text{ min}^{-1}$